

# Heitor Baldo, PhD

Applied Mathematician / Researcher

📞 (+55) 19 99081-9332  
✉ heitorbaldo@gmail.com  
🌐 heitorbaldo.github.io  
in heitorbaldo

## Academic Appointments

- 2024 – 2026 **Postdoctoral Research Fellow**, *University of São Paulo*, Brazil.
- 2024 – 2025 **Visiting Postdoctoral Research Fellow**, *Leipzig University*, Germany.
- 2019 – **Affiliate Researcher**, *IGDORE Institute*, Sweden.

## Education

- 2019 – 2024 **Ph.D. in Bioinformatics (Mathematical Neuroscience)**, *University of São Paulo*, Brazil.  
Thesis: *Towards a Quantitative Theory of Digraph-Based Complexes and its Applications in Brain Network Analysis*, advised by Koichi Sameshima and co-advised by André Fujita.
- 2014 – 2016 **M.S. in Applied and Computational Mathematics**, *University of Campinas*, Brazil.  
Thesis: *Álgebras de Clifford e de Cayley-Dickson*, advised by Jayme Vaz Jr.
- 2009 – 2013 **B.S. in Mathematics**, *University of Campinas*, Brazil.

## Publications

### Preprint Manuscripts

- 2026 **Baldo, H., Baccalá, L., Fujita, A., & Sameshima, K.**, *Directed Q-Analysis and Directed Higher-Order Connectivity on Digraphs: A Quantitative Approach*, arXiv:2605.14178, <https://doi.org/10.48550/arXiv.2605.14178>  
(preprint - submitted to a peer-reviewed journal)

### Under Review Manuscripts

- 2026 **Baldo, H. & Fujita, A.**, *A Temporal Hypergraph Motif Framework for the Analysis of Higher-Order Dynamics of Epileptic Brain Networks*, (under review).

### Unpublished Manuscripts and Academic Writings

- 2023 **Baldo, H.**, *Notes on Multi-Agent Reinforcement Learning*, (study notes).
- 2022 **Baldo, H.**, *Notes on Discrete Morse Theory on Digraphs*, (study notes).
- 2022 **Baldo, H.**, *Topics in Clifford and Lie Algebras associated with Graphs*, (unpublished Manuscript).

## Open-Source Software

Since 2022 **Research Packages:** DigplexQ, TemporalHypermotifs, PyTropical, Pytroids, Python, Sphinx, PyPi, NumPy, SciPy, Pandas, NetworkX, HyperNetX, XGI , <https://github.com/heitorbaldo?tab=repositories>

- Open-source Python packages providing computational tools for research in directed higher-order network analysis and directed Q-analysis, temporal hypergraphs, tropical mathematics, and graphic matroid theory.

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## Talks

- 2025 “Spectral Density of Multilayer Hypergraphs and Applications” in *6th Central German Meeting on Bioinformatics*, Leipzig University.
- 2022 “Path Homology Theory of Digraphs and Its Applications to Brain Network Analysis” in *Universit Leipzig - USP Workshop 2022* at the Institute of Mathematics and Statistics - USP.
- 2022 “Graph Cellular Automata and Beyond: Applications in Network Neuroscience.” Institute of Mathematics and Statistics - USP.

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## Teaching Experiences

- Spring 2021 **Teaching Assistant**, University of So Paulo, Brazil.  
Course: *Multivariate Data Analysis* (MAE0330).
- Winter 2017 **Lecturer**, University of Campinas, Brazil.  
Course: *Complex-Valued Neural Networks* (Minicourse).
- Spring 2016 **Teaching Assistant**, University of Campinas, Brazil.  
Course: *Linear Algebra* (MA327).
- Winter 2016 **Lecturer**, University of Campinas, Brazil.  
Course: *Bio-Inspired Algorithms: An Introduction* (Minicourse).

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## Technical Skills

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|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming    | Python • Julia • R • C/C++ • Rust • SQL • TypeScript • HTML/CSS                                                                                                     |
| ML/DL          | PyTorch • TensorFlow • scikit-learn • XGBoost • PyTorch Geometric • DGL                                                                                             |
| Backend & APIs | FastAPI • Pydantic • Uvicorn • PostgreSQL • Redis • SQLAlchemy • REST                                                                                               |
| Frontend       | Streamlit • Gradio • React • Next.js • TypeScript                                                                                                                   |
| MLOps & Infra  | Docker • MLflow • Weights & Biases • DVC • Evidently AI • Prometheus • Grafana                                                                                      |
| Math & Neuro   | SageMath • Mathematica • MATLAB/Octave • NetworkX/HyperNetX/XGI • Gudhi • Giotto-TDA • EEGLAB • Brainstorm • FreeSurfer • SPM • FieldTrip • fMRIPrep • Brian2 • MNE |
| Others         | Git • Linux • pytest • Sphinx • L <sup>A</sup> T <sub>E</sub> X                                                                                                     |

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## Selected Participation in Events

- 2026 Homotopy Theory, K-theory, and Topological Data Analysis. *Fields Institute*.
- 2025 Thirteenth Symposium on Compositional Structures (SYCO 13). *University of Cambridge*.

- 2025 Workshop “Biology and Geometry”. *Max Planck Institute for Mathematics in the Sciences.*
- 2025 6th Central German Meeting on Bioinformatics. *Leipzig University.*
- 2025 40th TBI Winterseminar in Bled. *Bled, Slovenia.*
- 2024 Twelfth Symposium on Compositional Structures (SYCO 12). *University of Cambridge.*
- 2021 Workshop on Algebraic Graph Theory and Quantum Information. *Fields Institute.*
- 2021 4th Workshop on Algebraic Graph Theory and its Applications. *Mathematical Center in Akademgorodok.*
- 2020 XLIII Annual Meeting of the Brazilian Society for Neuroscience and Behavior. *Online.*
- 2020 Seminars on Probability and Stochastic Processes. *University of São Paulo.*
- 2017 6th Brazilian Conference on Intelligent Systems. *Federal University of Uberlandia.*
- 2016 II Brazilian Congress of Young Researchers in Pure and Applied Mathematics. *University of Campinas.*
- 2015 IV School and Workshop on Lie Theory. *University of Campinas.*
- 2014 Workshop Many Faces of Distances. *University of Campinas.*

## Certifications & Training

- Spring 2026 Introduction to Agent-Based Modeling. *Santa Fe Institute.* [Certification]
- Spring 2018 Statistical Learning. *Stanford Online – Stanford University.* [Certification]
- Spring 2017 Introduction to Complexity. *Santa Fe Institute.* [Certification]
- Spring 2017 Minicourse on Machine Learning for Many-Body Physics. *IFT-Unesp.*

## Languages

- **Portuguese:** Mother Tongue.
  - **English:** Advanced Level (C1\*).
  - **German:** Pre-intermediate Level (A2\*)
- (\*) *Common European Framework of Reference (CEFR) level.*

Last updated August 02, 2026.